

Evaluating Explainable AI

From Quality Properties to Metrics

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A provocation

“SHAP is better than LIME.”

Better according to what?

Faithfulness? Stability? Compactness? Human usefulness?

Predictive performance is not explanation quality

Does f predict correctly? $\neq$ Does E describe f correctly?

- Predictive accuracy compares model outputs with target labels.
- Faithfulness concerns the explanation relative to model behavior.

What is being evaluated?

$$x \xrightarrow{f} f(x) \xrightarrow{E} e \quad Q(e, f, x, D, \theta_Q)$$

x input instance being explained

D data or background sample

f predictive model

$f(x)$ model output for x

E explanation method

e explanation produced by E

Q evaluation procedure

θ_Q metric/protocol parameters

We evaluate an explanation under a protocol, not in isolation.

A good explanation is not one thing

- Faithful but difficult to understand.
- Plausible but unfaithful to the model.
- Compact but incomplete.
- Stable but irrelevant to the user's task.

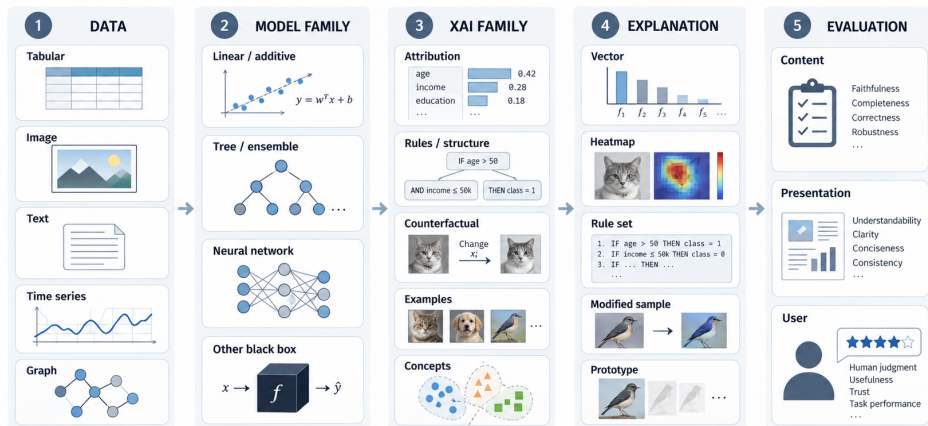
The scientific claim determines which trade-offs matter.

Data modality changes the evaluation operation

- **Tabular**: replace a feature or feature group.
- **Image**: mask a region or superpixel.
- **Text**: remove or replace a token.
- **Time series**: perturb an interval.
- **Graph**: remove a node or edge.

The same property can require different operational metrics for different representations.

A schematic XAI landscape



Content



Presentation



User

Co-12 instantiates the evaluation branch: what the explanation claims, how it is presented, and who uses it.

Three axes of an evaluation

Evaluation perspective

Where evidence comes from

Functionally grounded
Human-grounded
Application-grounded

Study design

How evidence is collected

Perturbation test
User simulation task
Expert decision study

Quality perspective

Which claim is tested

Content
Presentation
User

Example: a human-grounded simulation task can test a Content claim about faithfulness.

Keep these choices explicit: evidence source, procedure, and quality claim.

The literature uses overlapping language

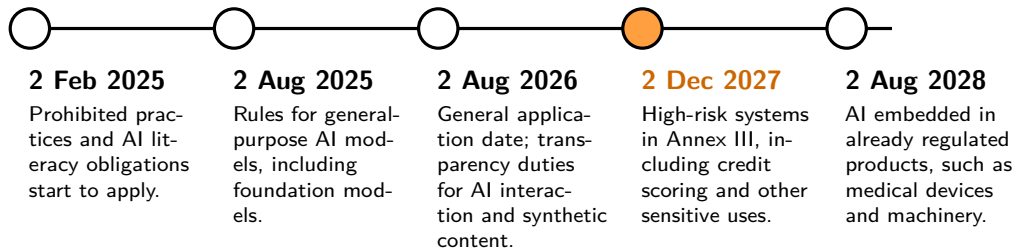
Explainability, interpretability, transparency, faithfulness, stability, robustness, comprehensibility, and actionability are related but not interchangeable.

Regulatory vocabulary: EU AI Act

The AI Act frequently uses **transparency**, **information to deployers**, **interpretation of outputs**, and human oversight language. In Recital 27, transparency is linked to traceability and explainability; this is related to XAI, but not the same as a metric definition.

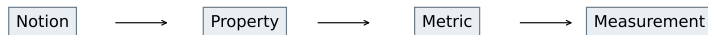
A shared vocabulary makes claims testable.

EU AI Act: application timeline



Sources: Regulation (EU) 2024/1689; Digital Omnibus on AI, Regulation (EU) 2026/1744, in force since 27 July 2026.

Four levels



Level	Meaning	Examples
Notion	Broad idea	transparency, explainability, interpretability
Property	Desired quality	faithfulness, stability, compactness, coherence
Metric	Operational test	deletion AUC, AOPC, max-sensitivity, entropy
Measurement	Protocol-bound value	deletion AUC = 0.31 with median baseline, seed 42

From property to measurement

Claim

The explanation is faithful to the predictive model.

One operationalization

Deletion AUC; replace top-ranked features by the training median.

Measurement

0.31 on a named model, dataset, target, baseline, and seed.

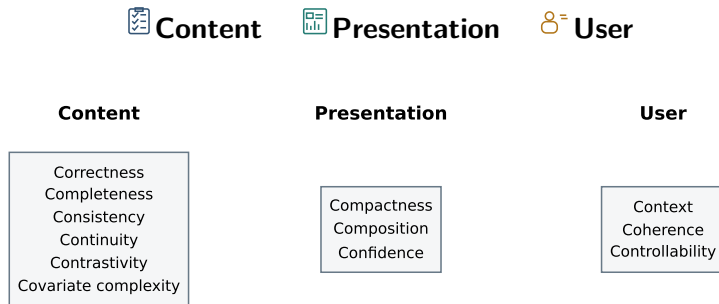
A scalar without its protocol is incomplete evidence.

The evaluation chain

Goal $\rightarrow$ Claim $\rightarrow$ Property $\rightarrow$ Metric $\rightarrow$ Protocol $\rightarrow$ Evidence

The metric is selected after the claim and property, not before them.

Co-12: a quality map, not a scorecard



Co-12 is a concrete quality map for the evaluation branch of the XAI landscape.[3]

Co-12: questions to ask

 Content	 Presentation	 User
Correctness / faithfulness	Compactness: concise enough?	Context: useful for this task?
Completeness	Composition: appropriate form?	Coherence: aligns with domain knowledge?
Consistency	Confidence: uncertainty represented?	Controllability: user can interact?
Continuity		
Contrastivity		
Covariate complexity		

One property, multiple metrics



Content Correctness / faithfulness

Deletion	Faithfulness correlation	Infidelity
Insertion	Randomization checks	Synthetic / white-box checks

Different operationalizations expose different assumptions and failure modes.[2]

Properties can conflict



Content: a faithful explanation may be verbose.



Presentation: a compact explanation may omit relevant behavior.

The evaluation objective must state which trade-offs are acceptable.

Faithfulness: perturb important features

Content

Correctness / faithfulness

Let $a = E(f, x)$, where a_i is feature importance. A perturbation-based test asks whether changing features with large $|a_i|$ changes $f(x)$.

$$\Delta_i = f(x) - f(x_{\setminus i})$$

The replacement rule is part of the metric.

Faithfulness correlation

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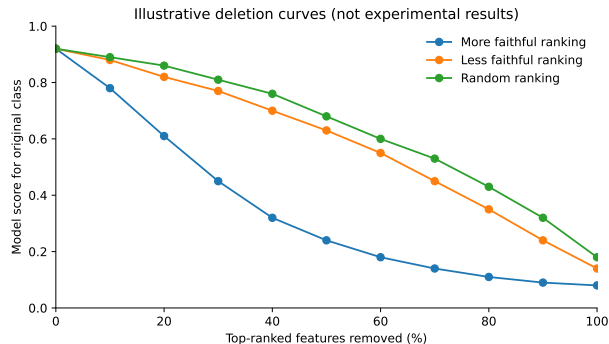
Correctness / faithfulness

$$\text{FC}(x) = \text{corr}_S \left(\sum_{i \in S} a_i, f(x) - f(x_{\setminus S}) \right)$$

Positive association is evidence that attribution mass tracks output change under the sampled subsets and replacement protocol.

Deletion curve

Rank features by $|a_i|$, then progressively replace the most important ones. A steep score decrease early in the curve is consistent with a useful ranking under this deletion rule.



Deletion AUC and AOPC

$$\text{AUC}_{del} = \int_0^1 f(x^{(\alpha)}) d\alpha \quad \text{AOPC} = \frac{1}{K} \sum_{k=1}^K [f(x) - f(x^{(k)})]$$

For a score-preserving output and a fixed deletion rule, lower deletion AUC or higher AOPC is generally preferred. Always state direction and target output.

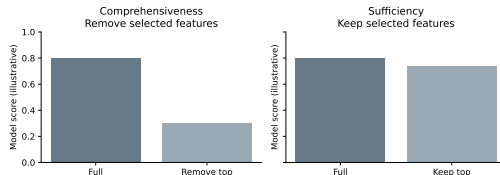
Comprehensiveness and sufficiency

Content

Completeness

$$\text{Comp}(x, S) = f(x) - f(x_{\setminus S}) \quad \text{Suff}(x, S) = |f(x) - f(x_S)|$$

- Remove selected features: does the score change? Higher drop is generally preferred.
- Keep selected features: does the score remain? Smaller difference is generally preferred.



Infidelity

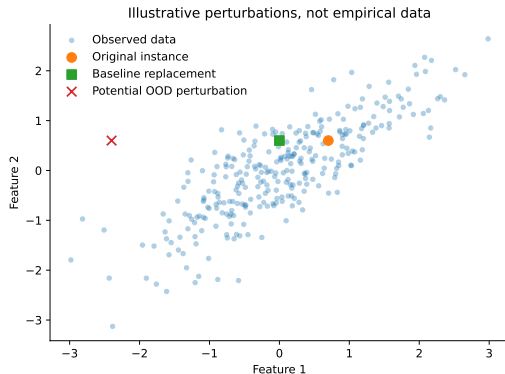
Content

Correctness / faithfulness

$$\text{Infid}(E, f, x) = \mathbb{E}_I \left[(I^T E(f, x) - (f(x) - f(x - I)))^2 \right]$$

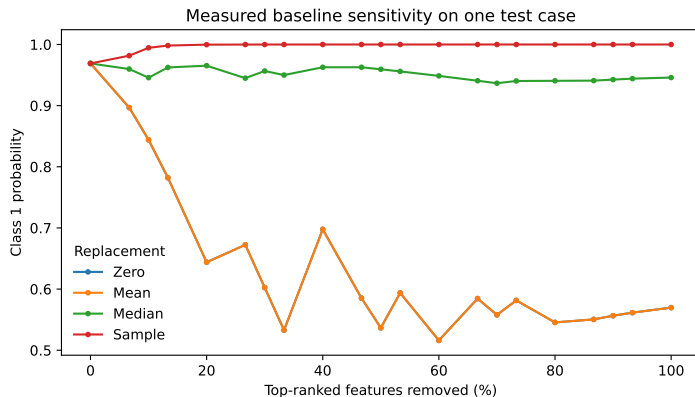
It compares attribution-predicted change with actual model change. The perturbation distribution I defines what changes are tested.[4]

The perturbation problem



- What does “feature absence” mean: zero, mean, median, or conditional sample?
- If replacement leaves the data manifold, are we measuring the explanation or model extrapolation?

Baseline sensitivity is a result



Measured on one held-out instance; the curve changes when the replacement convention changes.

Consistency is not continuity



Content

Consistency and continuity

Consistency Equivalent input/model conditions yield equivalent explanations.

Continuity Similar inputs with similar predictions yield similar explanations.

Repeated seeds and equivalent implementations probe consistency; nearby inputs probe continuity.

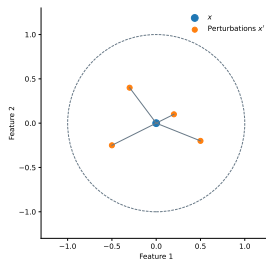
Max-sensitivity

Content

Continuity

$$\text{Sens}_{max} = \max_{\|x' - x\| \leq r} \|E(f, x) - E(f, x')\|$$

The radius, input norm, explanation distance, and validity of perturbed inputs all matter.



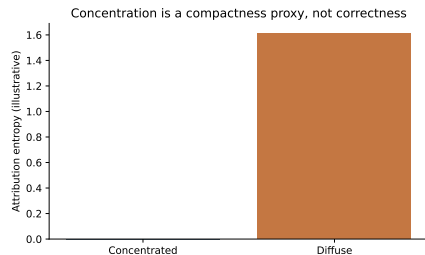
Compactness is not correctness

 **Presentation** Compactness

 **Content** Correctness

$$a = [0.91, 0.04, 0.03, 0.01, 0.01] \quad b = [0.21, 0.20, 0.20, 0.20, 0.19]$$

Which is easier to inspect? Does that make it more faithful?



Entropy and sparsity

Presentation

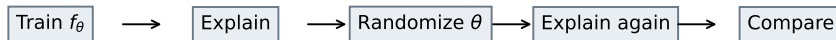
Compactness

Normalize $p_i = |a_i| / \sum_j |a_j|$ and compute

$$H(a) = - \sum_i p_i \log p_i.$$

Lower entropy indicates concentration; higher Gini indicates concentration. Neither establishes that the explanation is correct or useful.

Sanity checks



If explanations barely change after model parameter randomization, model dependence is questionable. Passing a sanity check is necessary evidence, not proof of correctness.[1]

Faithful does not imply understandable

 **Content**

 **Presentation**

Faithful $\nRightarrow$ Understandable

A technically faithful explanation with a large number of predicates may be unusable for its audience.

Plausible does not imply faithful

 **User**

 **Content**

Coherent with human knowledge $\nrightarrow$ Correct relative to model

A domain expert may find an explanation plausible even if the model relies on a spurious feature.

Human-grounded tasks

 **User**

Coherence, context, controllability

- Predict the model output; detect a model failure.
- Identify influential variables; answer comprehension questions.
- Measure decision time, confidence, and calibration.

Preference alone is weak evidence: “Which explanation do you like?” does not test task performance.

Application-grounded questions

 **User**

Context and task performance

Physician Does the explanation support diagnosis or error detection?

Engineer Does it help debug model behavior?

Auditor Does it reveal unacceptable dependencies?

Scientist Does it generate useful hypotheses?

A reproducible teaching benchmark

- Breast Cancer Wisconsin data; training-only standardization.
- Logistic regression, random forest, and MLP; class-1 probability.
- Local replacement attribution; model-specific attribution where available; random control.
- Median baseline; 25 selected test cases; explicit perturbation radius.

Designed for a short class run, not state-of-the-art comparison.

Metric matrix: keep directions visible



Content Faithfulness, completeness, continuity



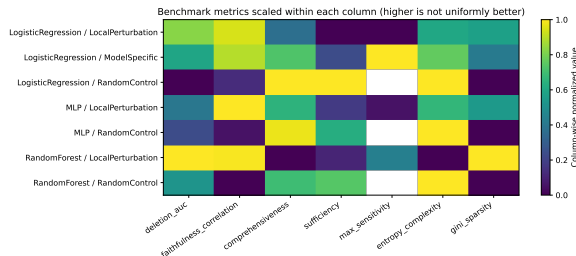
Presentation Compactness

Deletion AUC ↓ Faithfulness correlation ↑

Comprehensiveness ↑ Sufficiency ↓

Max-sensitivity ↓ Gini sparsity: descriptive only

The table and plots below are computed from per-instance benchmark outputs. Column scaling is for visual comparison, not a common utility score.



Include negative controls



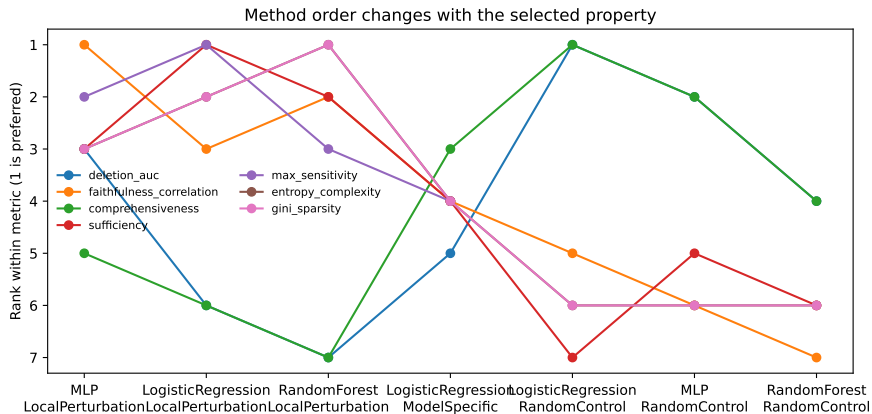
Content

Correctness / faithfulness

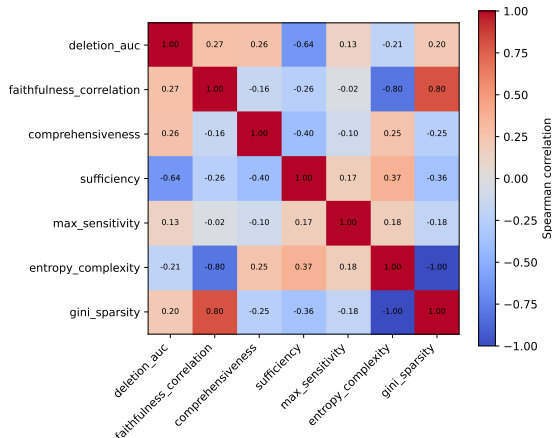
Random feature rankings and model randomization ask whether the protocol can reject explanations that should not earn high faithfulness.

An evaluation without controls can produce precise-looking but uncalibrated scores.

Metric disagreement

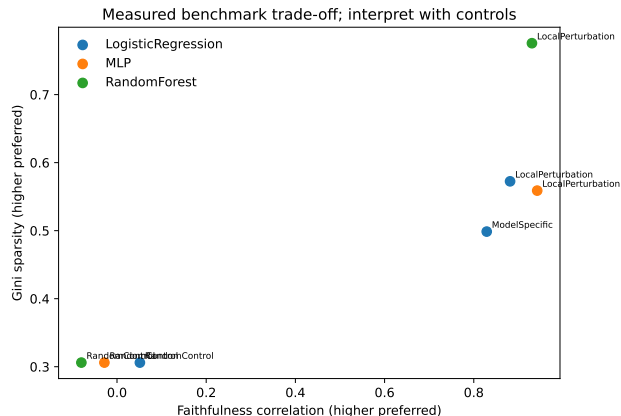


Metrics do not necessarily agree



Ranks and correlations describe this benchmark's models, samples, and protocol. They do not define a universal ordering.

Multi-objective view



Content faithfulness, stability



Presentation compactness

Avoid aggregation until the application defines a utility.

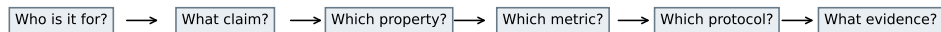
Discussion

 **Content**  **Presentation**  **User**

- Better deletion AUC but worse stability: which method is better?
- If deletion is out of distribution, what did the score test?
- What control should a benchmark be able to reject?

Answer by naming the claim and intended use.

A defensible evaluation workflow



Then specify baselines, controls, aggregation, uncertainty, and the need for human evaluation.

Take-home points

 **Content**  **Presentation**  **User**

1. Explanation quality is multidimensional.
2. A property and a metric are not the same thing.
3. Metrics encode assumptions about perturbations, baselines, and outputs.
4. Technical validity and human usefulness are different questions.
5. Strong evaluation combines complementary evidence and controls.

Final question

Do not evaluate an explanation.

Evaluate claims about an explanation.

Which claim? Which property? Which metric? Which protocol? Compared with what control?

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